Arming, Alliance, and Screening

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Motivation: Alliances

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Literature: Alliances can deter war through costly signaling (Morrow 1994; Smith 1995)

$\leftarrow$ solving information problems on a patron's intention.

But there is little discussion on commitment problems and preventive war.

Are alliances effective against another cause of war?

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Preventive war is

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Counter examples: Japan and West Germany after WWII

Why did preventive war fail to occur in these likely cases?
How do alliances work under power shifts?

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RQ: Do alliances deter preventive war? If so, how?

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My argument

A screening effect of alliances: Alliances make arming more informative, and this screens the otherwise hidden intention of a rising protégé.

Model: Setup

Overview

In the baseline model,

- Two actors (A and B)
- a state experiencing resource growth (A) chooses whether to invest in its military capability
- Expecting a future power shift, a declining state (B) wants to prevent it if A is a revisionist, but wants to pass if A is a pacifist
- revisionism (r) is private information

In the main model, an alliance is introduced.

- Three actors (A, B, and E)
- A has a defensive alliance with E
- E is SQ oriented, and can intervene in a war

Model: Setup

Sequence

- *Nature* chooses A's revisionism ($r \in \{1, p\}$, $0 < p < 1$). B and E do not know it.
- A chooses whether to arm at costs ($K = k > 0$) or not ($K = 0$)
- B chooses whether to launch a preventive attack on A before arming is complete ($a = 0$)
- If B passes, A's revisionism is revealed and arming is complete ($a = 1$)
- B offers $x \in (0, 1)$ in crisis bargaining, and A decides whether to accept ($\rightarrow$ settlement) or reject it ($\rightarrow$ war)
- E is SQ oriented and can intervene in a war, increasing A's power by ϵ

Model: Payoffs

General war payoffs: $rw_A(\cdot)$ for A and $w_B(\cdot)$ for B.

- $w_A(\cdot)$ is increasing and $w_B(\cdot)$ is decreasing in the input $(\cdot)$
- War is inefficient: $w_A(\cdot) + w_B(\cdot) < 1$
- the revisionist type $(r = 1)$ gets a larger war payoff than the pacifist type $(r = p)$ $(w_A(\cdot) > pw_A(\cdot))$

E.g.,

Preventive war $(a = 0)$ w/o intervention: $w_A(0)$ and $w_B(0)$

War after arms buildup $(a = 1)$ w/o intervention: $w_A(1)$ and $w_B(1)$

War after arms buildup w/ intervention: $w_A(1 + \epsilon)$ and $w_B(1 + \epsilon)$

Model: Payoffs

$$u_i(\text{Settlement}) = \begin{cases} x - K & (\text{if } i = A) \\ 1 - x & (\text{if } i = B) \\ 1 - |x - s| & (\text{if } i = E) \end{cases} \tag{1}$$

$$u_i(\text{War w/o Intervention}) = \begin{cases} rw_A(a) - K & (\text{if } i = A) \\ w_B(a) & (\text{if } i = B) \\ 1 - |rw_A(a) - s| & (\text{if } i = E) \end{cases} \tag{2}$$

$$u_i(\text{War w/ Intervention}) = \begin{cases} rw_A(a + \epsilon) - K & (\text{if } i = A) \\ w_B(a + \epsilon) & (\text{if } i = B) \\ 1 - |rw_A(a + \epsilon) - s| & (\text{if } i = E) \end{cases} \tag{3}$$

, where s is an ideal point for E.

Assume E intervenes in a preventive war, but not in wars after arming is complete.

Implication

A *screening* effect of alliance: alliances make arming more informative by changing the value of arming

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- Both types of A want to build their arms
  $\rightarrow$ the inability to distinguish types
  $\rightarrow$ the revisionist type mimics the pacifist type
  $\rightarrow$ preventive war occurs with some probability ($m^* = \frac{p(w_A(1)-w_A(0))-k}{p(w_A(1)-w_A(0))}$)

, which is consistent with Debs and Monteiro (2014)

Implication

In the main model,

- The pacifist A does not need to arm thanks to the alliance
- Only revisionist A has the incentive to arm
  $\rightarrow$ arming is a clear sign of revisionism
  $\rightarrow$ arming *would* cause preventive war *for sure*
- Given this, the revisionist is *less* likely to arm to avoid a preventive war
  $\rightarrow$ preventive war is less likely
  $\rightarrow$ screening happens in an off-equilibrium path

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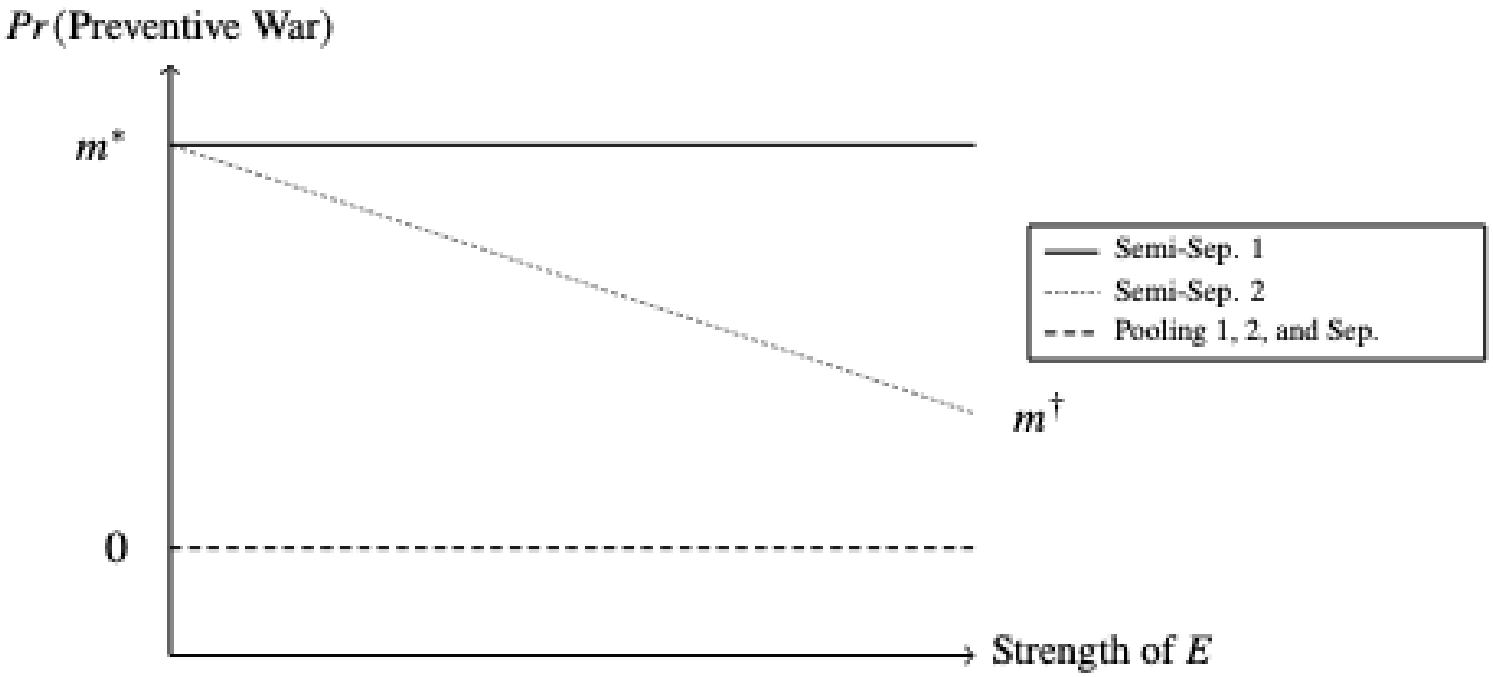
In this screening effect, alliances can simultaneously

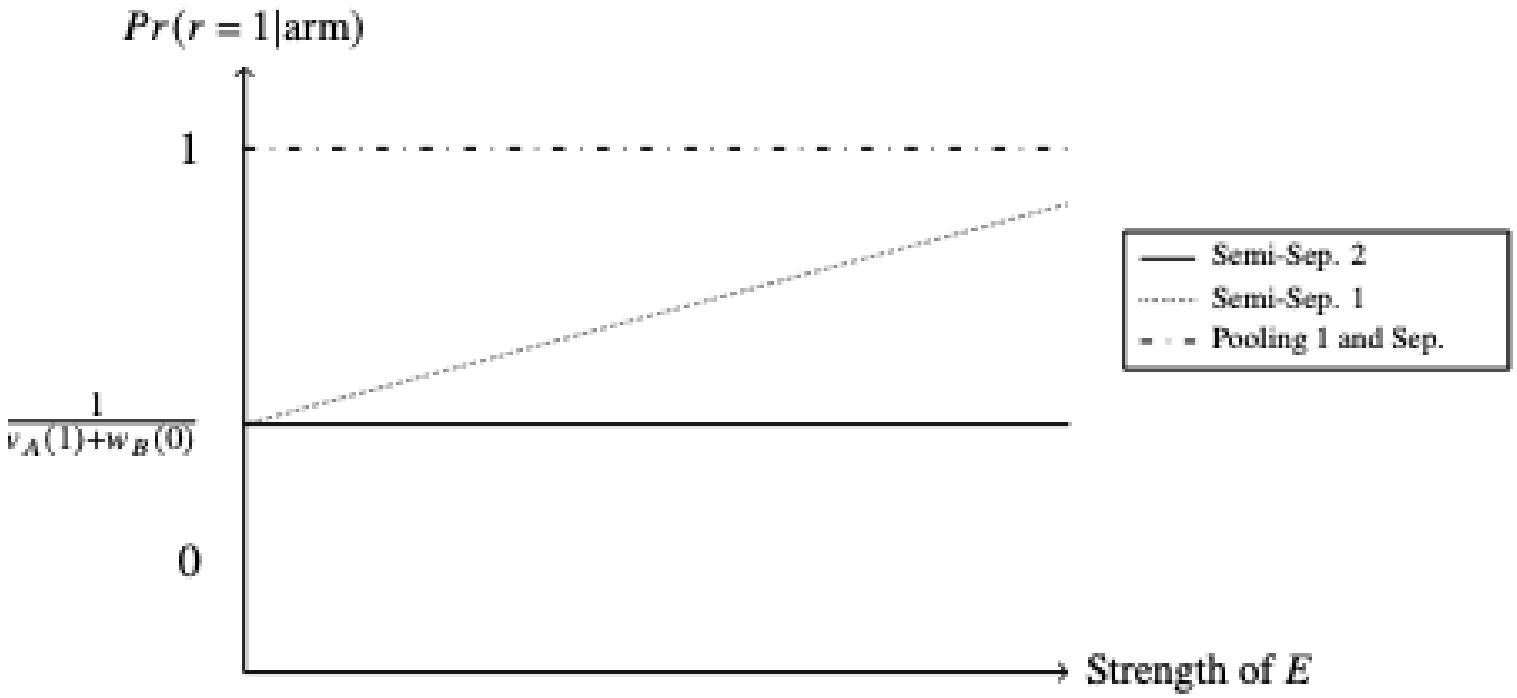
- Deter a preventive attack
- Constrain the arming by a protégé
- In a different way from existing studies (Benson, 2012; Fang et al., 2014)

Implication: Probability of Preventive War



Pr(Preventive War) is higher in the baseline alliance model (solid line) and lower in the main alliance model (dotted lines)

B's Belief and Screening Effect



B's belief that A is a revisionist after arming ($Pr(r = 1|\text{arm})$) is higher in the main model (dotted lines) than the baseline model (solid line).
$\rightarrow$ arming, coupled with an alliance, is a clear sign of revisionism.
$\rightarrow$ a sign of the screening effect

Hypothesis and Empirical Strategy

- **Hypothesis**: Defensive alliances decrease the probability of preventive war

- I follow Bell and Johnson (2015) and recalculate the expected power shifts
- DV: War on target (a rising side)
- IV: Expected power shifts, defense pacts[1], and their interaction term
- with some pre-treatment control variables[2] and directed-dyad- and year-fixed effects, estimated through OLS[3]

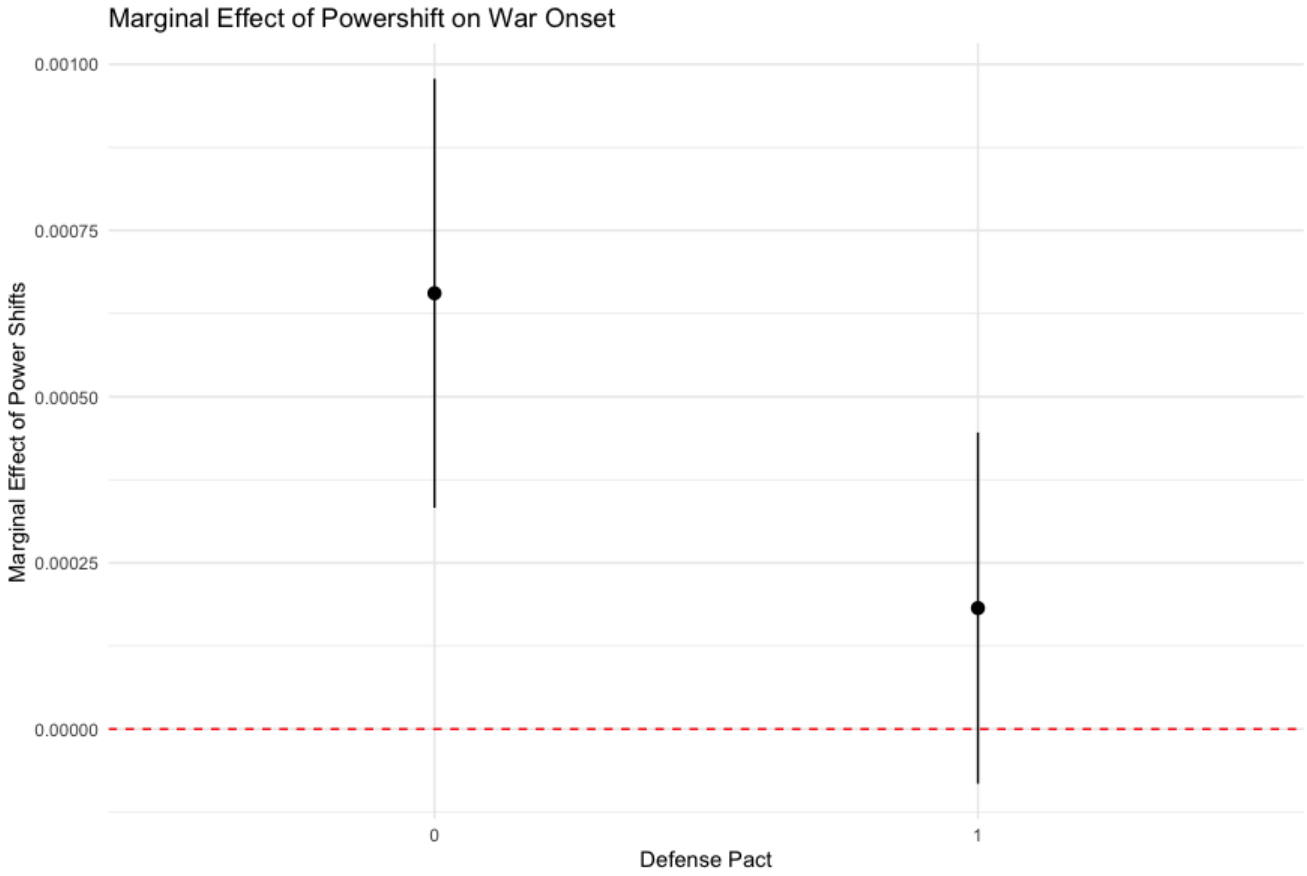
$$Y_{it} = \beta_1 \text{PS}_{it} + \beta_2 \text{Alliance}_{it} + \beta_3 \text{PS}_{i,t} \times \text{Alliance}_{i,t} + \mathbf{X}'\beta + \lambda_i + \eta_t + \epsilon$$

[1] From ATOP dataset.

[2] Mutual democracy (Polity2), foreign policy similarity (S-score), contiguity, distance, and up to a cubic polynomial of peace years

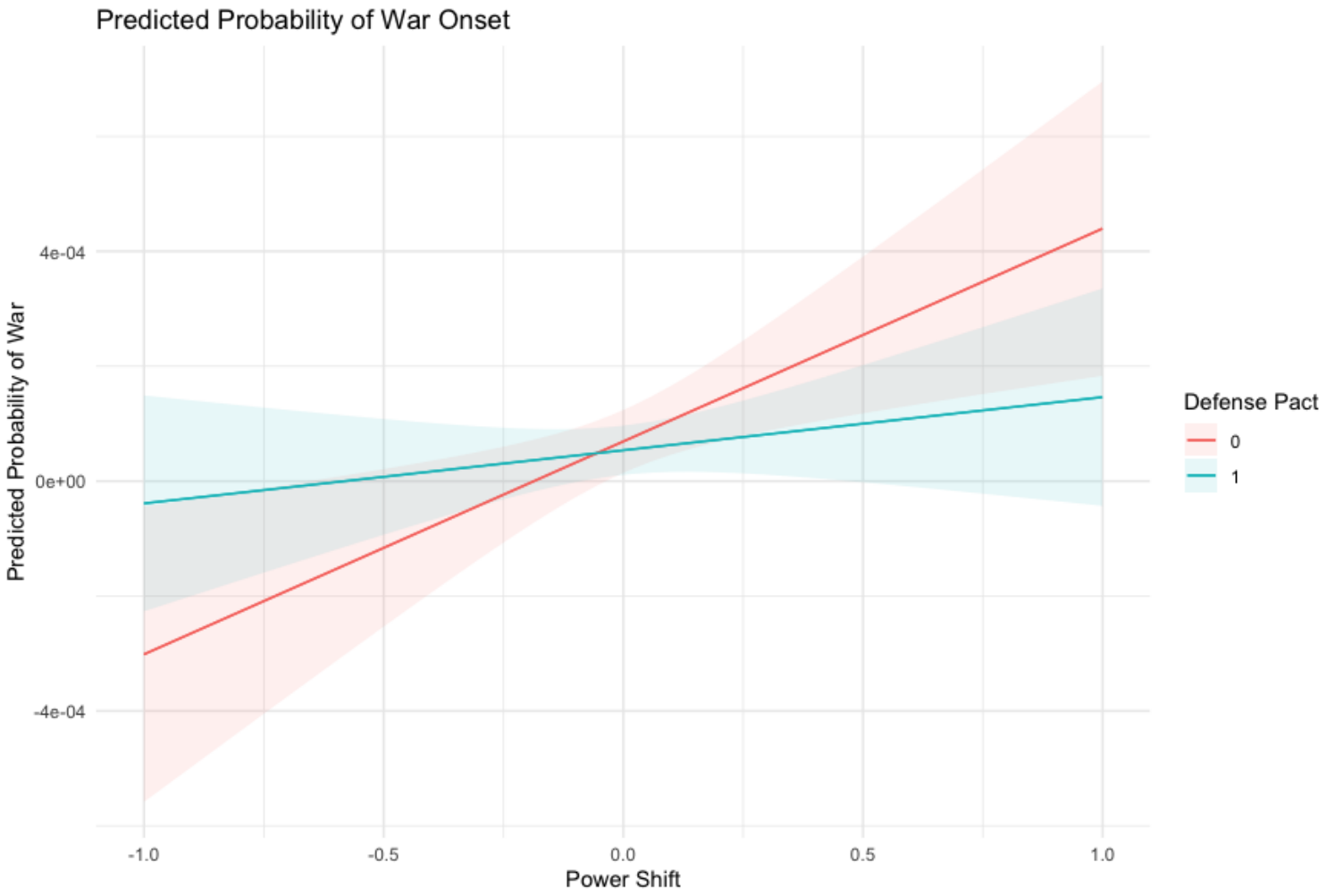
[3] See Beck (2020) for methodological concerns of logistic regression with fixed effects.

Result: Marginal Effect of Expected Power Shifts



- the expected power shifts are positively associated with war
- alliances mitigate this positive correlation

Result: Predicted Probability of War



Robustness

These results remain robust across multiple model specifications.

- a different foreign policy similarity measure (UN Voting Data)
- a different democracy index (Vdem)
- excluding NATO
- different year coverage without NATO (before and after 1945)
- logistic regressions

Case Studies

Two case studies to test the mechanisms.

Germany in the July Crisis in 1914

- The Franco-Russian Alliance
- Russian general mobilization on July 31st
- Germany's preventive attack on August 4th
- Thanks to the alliance, Germany was convinced that Russia was a revisionist state

Japan after WWII

- Clear understanding that arming would destabilize the region
- Low military expenditure and efforts to reconstruct relationships
- Nonetheless, enduring distrust especially from South Korea and China
- because, due to the screening effect, a revisionist Japan would also do the same thing

Conclusion

This paper

- Investigates whether alliances deter preventive war
- Proposes a *screening effect* of alliances with some qualitative and quantitative evidence
- Suggests that alliances serve as a critical institutional tool for sustaining global stability during periods of power shifts

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Thank you for listening!

Appendix: Equilibrium

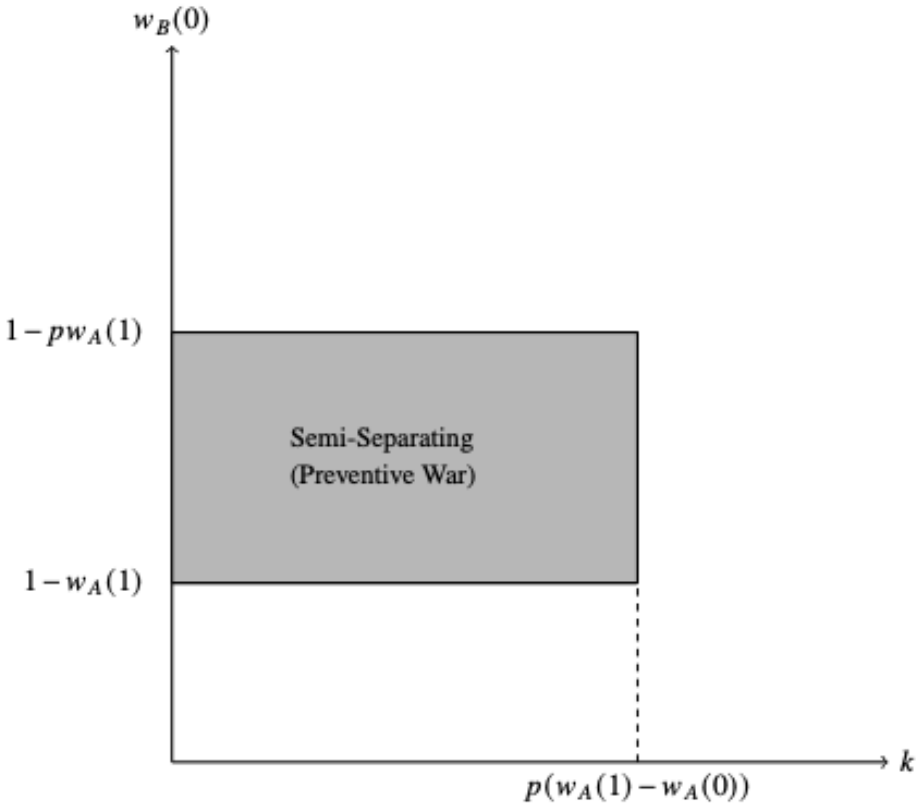

Figure: Equilibrium without Alliance

Appendix: Equilibrium

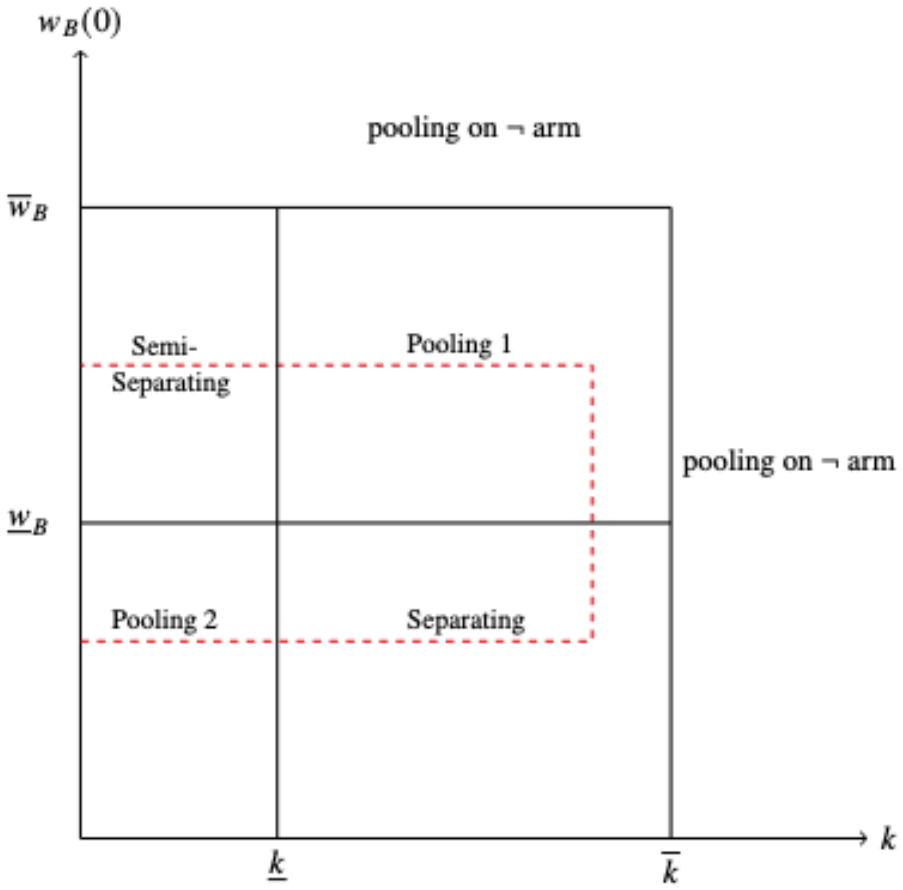

Figure: Equilibrium with Alliance

Appendix: Equilibrium

Proposition 1 (Semi-Separating). *When Line 6 is satisfied and*

$$p(w_A(1) - w_A(0)) > k \tag{7}$$

$$\phi < \phi^* = \frac{1 - pw_A(1) - w_B(0)}{w_A(1)(1-p)}, \tag{8}$$

there exists a semi-separating PBE at which Revisionist A always arms, SQ type A arm with $l^* = \frac{\phi(w_A(1)+w_B(0)-1)}{(1-\phi)(1-pw_A(1)-w_B(0))}$, *B attack A with* $m^* = \frac{p(w_A(1)-w_A(0))-k}{p(w_A(1)-w_A(0)}$ *after arming, and does not attack after no arming. See Appendix for proof.*